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Consolidation of episodic memory: an epiphenomenon of semantic learning

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Abstract

Two hypotheses dominate the literature on systems consolidation of episodic memory: the transfer hypothesis and the transformation hypothesis. The former postulates a transfer of episodic memory from a fast-learning hippocampus to a slow-learning neocortex. The latter postulates that only the hippocampus genuinely stores episodic memories, and that systems consolidation arises due to multiple memory traces in the hippocampus and a transformation of episodic into semantic memories. While both hypotheses are supported by some evidence, they are contradicted by other, and hence remain controversial. Here, I suggest a new account of systems consolidation. It is based on the transformation hypothesis and introduces two modifications. First, episodic and semantic memory differ in their representational format, which is optimized for different purposes: Rigid sequences for episodic memories and flexible representations for semantic memory. Second, multiple memory traces in the hippocampus are not required to account for the temporal gradient of retrograde amnesia, if there is any. To this end, slow semantic learning from episodic memories suffices. The main hypothesis that I propose in this chapter is that systems consolidation is an epiphenomenon of semantic learning from episodic memory.

Introduction

Müller and Pilzecker (1900) coined the term “consolidation” more than 100 years ago, but we still do not fully understand why consolidation occurs and what happens to memories during this process. In the current literature, consolidation might refer to several different phenomena (see also Chapter by Genzel and Wixted). Synaptic consolidation refers to

protein-synthesis-dependent late-phase long-term potentiation (Frey and Morris, 1997; Bramham and Messaoudi, 2005; Dudai, 2012) and consolidation in procedural learning refers to a stabilization of motor memory (Brashers-Krug et al., 1996) or an improvement in performance during a rest period post-training (Walker et al., 2002). The contemporary term “systems consolidation” most closely matches the phenomenon Müller and Pilzecker originally studied: The process by which an initially labile memory is made stable such that it becomes robust to interference or brain injury (Rosenbaum et al., 2008; Dudai, 2012). Here, we focus on the systems consolidation of episodic memory, the memory of personally experienced events. About 60 years ago, it was discovered that episodic memories cannot be stored without a functioning hippocampus (anterograde amnesia) (Scoville and Milner, 1957). For memories that were stored before the hippocampal insult, the impact is less severe on remote memories than on more recent memories (graded retrograde amnesia) (Zola-Morgan and Squire, 1990; Kim and Fanselow, 1992; Rosenbaum et al., 2008).

Theoretical accounts of consolidation: transfer vs. transformation

Due to their influence on the field two hypotheses stand out from among the many hypotheses about systems consolidation. According to the *transfer hypothesis*, the memory contents are transferred from the hippocampal memory system to a neocortical memory system during systems consolidation (Buzsaki, 1989; Alvarez and Squire, 1994; McClelland et al., 1995). Systems consolidation is necessary because a neural network cannot store memories both rapidly and stably over time because of catastrophic interference (McClelland et al., 1995). With two complementary learning systems (CLS), episodic memories can be stored quickly in the hippocampus, where plasticity is rapid, and then gradually transferred to the neocortex, where memories are encoded more slowly, but also more stably. The memory contents are hypothesized to remain similar, if not identical, during and after consolidation. The transfer hypothesis is supported, for instance, by findings in imaging studies that hippocampal activation during retrieval is lower for more remote than for recent memories (Bontempi et al., 1999; Takashima et al., 2009; see Chapters by Genzel and Wixted, by Fernandez and by Genzel and Battaglia).

However, there are challenges for the transfer hypothesis (Nadel and Moscovitch, 1997; Winocur and Moscovitch, 2011; see Chapter by Sekeres, Moscovitch and Winocur). For instance, there is currently little experimental support for a fundamental difference in the time scales of plasticity in the hippocampus and neocortex (Cheng, 2013). On the contrary, some studies find rapid neocortical learning on the timescale of single trials (Sharon et al., 2011). The most serious challenge, however, is the long temporal gradient of retrograde amnesia. In amnesics, whose brain damage was limited to the hippocampus, the gradient extends 15 years or more in some cases (Squire et al., 1989) and retrograde amnesia appears to be not graded at all in other cases (Steinvorth et al., 2005). It is difficult to conceive why the transfer process would last more than a decade or the entire lifetime of an individual. Moreover, such prolonged storage in the hippocampal memory system contradicts the core assumption of the transfer hypothesis that memory storage in the hippocampus is not stable (Buzsaki, 1989).

An alternative view of systems consolidation is the *transformation hypothesis* (Winocur and Moscovitch, 2011). It holds that genuine episodic memories are stored only in the hippocampus and that their recall requires the hippocampus regardless of the age of the memory (Nadel and Moscovitch, 1997). This prediction is supported by fMRI studies (Nadel et al., 2000; Ryan et al., 2001; Harand et al., 2012). Multiple Trace Theory (MTT) suggests that each retrieval gives rise to a new copy of the memory trace. So older episodic memories, which have been retrieved more frequently, will be represented by more memory traces than recent memories. Therefore, partial hippocampal lesions are less likely to affect remote than recent memories because remote memories have more widely distributed representations in the network. In short, MTT can account for the long (Squire et al., 1989) or flat gradient (Steinvorth et al., 2005) gradient in retrograde amnesia, which are difficult to reconcile with the transfer hypothesis.

The transformation hypothesis additionally holds that episodic memories are transformed into semantic memories (Cermak, 1984; Nadel and Moscovitch, 1998; Winocur et al., 2010; Winocur and Moscovitch, 2011). When episodic memories are reactivated, there are some deviations in details. Those central features that overlap in different memory traces, the gist,

can be identified by neocortex and stored as semantic memory¹. In other words, during consolidation there is a “loss of detailed, contextual features.” (Winocur and Moscovitch, 2011, p. 767). Such a process has been observed in animal experiments (Wiltgen and Silva, 2007). Once a semantic memory has been established, the same information is located in two different sites: once in the hippocampus and once in the neocortex. Hence, remote autobiographical information could be recalled from semantic memory in the neocortex, even if the hippocampus were completely lesioned. This would make it appear as if some episodic memory were stored outside the hippocampus.

A central element of the transformation hypothesis is the distinction between episodic memory and semantic memory, which the hypothesis relies on to account for the different functions of hippocampus and neocortex and to account for systems consolidation. We therefore review the evidence for a dissociation between episodic and semantic memory in the following.

The distinction between episodic and semantic memory

Semantic memory is thought to be the memory of general facts (Quillian, 1966; Collins and Quillian, 1969; Tulving, 1972), which differs from the memory of particular episodes in episodic memory. When Tulving (1972) first conceptualized episodic memory as memory of personally-experienced events, he hypothesized that episodic memories could be distinguished on the basis of the information that they contain. He suggested that “[e]pisodic memory receives and stores information about temporally dated episodes or events, and temporal-spatial relations among these events.” (Tulving, 1972, p. 385) This criterion later came to be known as the what-where-when (Griffiths et al., 1999) or WWW criterion (Suddendorf and Busby, 2003). Tulving (1972) thought that, on this basis, episodic memories

¹ Interestingly, in the most widely cited article that champions the transfer hypothesis, McClelland et al. (1995) propose that the hippocampus stores single experiences whereas the function of neocortex is “discovering structure”. These ideas are reminiscent of the transformation hypothesis, but the computational models in McClelland et al. assume that the contents of both episodic and semantic memory are very similar, if not identical.

could be distinguished from other memory systems such as semantic or procedural memory. However, Tulving (1985) later realized that the WWW criterion was neither necessary nor sufficient to define episodic memory. For instance, we frequently know what happened where and when (e.g., such as details about Abraham Lincoln's assassination) without having personally experienced the event. Tulving therefore developed a new definition that focused on the subjective experience during the retrieval of episodic memories. He proposed that anoetic, noetic, and auto-noetic awareness characterize the recall of procedural, semantic and episodic memory, respectively. Roughly speaking, anoetic consciousness lacks awareness of the content of the memory, noetic consciousness includes awareness of the content, and auto-noetic consciousness includes the awareness that one self has been part of the recalled event. The experience during retrieval of episodic memories was likened to mental time travel into the past (Tulving, 1985; Suddendorf and Corballis, 1997), a virtual reliving of the past.

Notwithstanding the lack of a clear definition between the two types of memory, numerous findings suggest that two separate memory systems exist. Patients with lesions of the medial temporal lobe (MTL) have a severe impairment of episodic memory (Bayley et al., 2008; Rosenbaum et al., 2008), but their semantic memory appears to be largely spared (Hirano and Noguchi, 1998; Schmolck et al., 2002) and they can learn new semantic information (Vargha-Khadem et al., 1997; Bayley et al., 2008; Sharon et al., 2011). By contrast, patients with semantic dementia suffer from a severe loss of semantic memory, while their episodic memory is relatively spared (Snowden et al., 1994; Hodges and Patterson, 2007). Further dissociations have been observed in neuroimaging studies. Semantic memory activates the left inferior prefrontal cortex and left posterior temporal areas (Wiggs et al., 1998), whereas autobiographical and episodic memory retrieval involves increased activity in the medial frontal cortex, middle temporal, temporopolar and right prefrontal areas (Maguire, 2001; Gilboa, 2004). Taken together, these findings suggest to some authors that episodic memory is a separate memory system from semantic memory.

However the above conclusion is not uncontroversial. Some authors have argued that the conclusion is based on studies that have focused only on activation difference where the

experimental paradigms and statistical methods were designed to dissociate the neural correlates of the two types of memories (Rajah and McIntosh, 2005; Burianova and Grady, 2007). Other studies, found that amnesics have semantic retrieval deficits in addition to profound impairments in episodic memory (Squire and Zola, 1998; Kopelman and Kapur, 2001; Manns et al., 2003), suggesting that the MTL is important for both semantic and episodic retrieval. Furthermore, the left prefrontal cortex, which is thought to be part of the semantic memory system, is also involved during episodic memory retrieval (Martin and Chao, 2001; Wagner et al., 2001). Damage to left prefrontal cortex has been found to impair free recall and context memory task performance (Stuss et al., 1996; Mangels, 1997).

As a result, alternative views have emerged. The unitary system view suggests that a single declarative memory system subserves both episodic and semantic memory (Baddeley, 1984; Rajah and McIntosh, 2005; Burianova and Grady, 2007). Specifically, the encoding of to-be-remembered material is always contextual (Baddeley, 1984), but at retrieval, memories may, or may not, become decontextualized depending on the task demands (Westmacott and Moscovitch, 2003; Rajah and McIntosh, 2005; Yassa and Reagh, 2013). What is called episodic memory, they posit, is simply a memory that is more contextual than one that is called semantic. Indeed, semantic memory is rarely entirely context-free, but rather may involve some contextual and episodic components (Westmacott and Moscovitch, 2003; Gilboa, 2004), and episodic retrieval is not free of general world knowledge (Levine et al., 2002; Gilboa, 2004). Similarly, Mayes and Roberts (Mayes and Roberts, 2001) suggest that episodic memory is the result of a synergy between perceptual and semantic aspects of objects and spatio-temporal contextual features. In a similar vein, Klein (2013) proposed that one and the same memory trace could be retrieved as either episodic or semantic memory, depending on whether the retrieval was associated with auto-noetic or noetic consciousness, respectively.

Greenberg et al. (2010) suggested that memory exists on a continuum, and that episodic and semantic memory merely represent two extremes of the same type of memory. As a consequence, intermediate manifestations of that memory also exist and episodic memory

cannot clearly be separated from semantic memory. A number of neuroimaging and neuropsychological studies reported commonalities in neural activations across tasks involving semantic and episodic memories (Rajah and McIntosh, 2005; Burianova and Grady, 2007). For example, in an event-related functional magnetic resonance imaging (fMRI) study, Burianova et al. (2010) showed subjects pictures to cue three types of memory retrieval: autobiographical, episodic, and semantic. In every retrieval condition, they found that three regions (left hippocampus, left lingual gyrus, and right caudate nucleus) showed a common pattern of connectivity. Along similar lines, the semantic system identified by Binder et al. (2009) in a meta analysis of 120 functional neuroimaging studies overlaps to a large extent with the core network of the episodic memory system (Maguire, 2001; Svoboda et al., 2006).

The component-structure of episodic memory

In the following, I suggest a novel account of systems consolidation, which largely adopts the transformation hypothesis, but deviates from it in two significant points. First, in contrast to previous approaches that were based on memory content or subjective experience, I suggest that episodic and semantic memory differ in their representational format. Each representational format is optimized for a different purpose. The sequential character of one-time experiences involving arbitrary objects and modalities requires that episodic memories are represented by neuronal sequences, whereas capturing propositions about particular objects and the relationships between them requires flexible representational structures in semantic memory. Second, multiple memory traces in hippocampus are not required to account for systems consolidation. Slow semantic learning from episodic memories alone can account for a temporal gradient of retrograde amnesia, if there is any. However, slow learning is not due to slow plasticity in neocortex as the transfer hypothesis would have it, but rather due to the nature of semantic information.

Perhaps episodic memory and its consolidation are so difficult to understand because episodic memory in humans is a complex compound phenomenon. If so, then the first step to

understanding its nature and neuronal basis is to dissociate episodic memory into its constitutive components. Adopting this view, I suggest here that the following components can be distinguished in episodic memory: the subjective experience, scenario generation, episodic memory traces², semantic representations, and semantic information. Except for the distinction between semantic representation and semantic information, these components were introduced previously (Cheng et al., 2016). For the sake of brevity, the subjective experience in episodic memory is not discussed here, since in my view it does not add to our understanding of systems consolidation. Nevertheless, the nature of the subjective experience during retrieval can be incorporated into the framework for episodic memory proposed here (Cheng et al., 2016).

Semantic representations and information

Semantic memory, as used in the field, is probably an umbrella term that subsumes several concepts that are quite different from each other (Renoult et al., 2012). For this reason, I use the more specific terms semantic representation and semantic information instead in the following.

Semantic representations are needed in an episodic memory to account for the finding that episodic memory is not detailed, but contains only gist information (Roediger and McDermott, 1995; Koutstaal and Schacter, 1997; Oliva, 2005). I regard the gist as the information content from the most abstract level that is relevant in that one episode. The relevant level is established dynamically for each episode depending on the context and attention (Graham et al., 2000). In some cases, the most abstract level could in fact be quite detailed, a level that in most other cases would not enter into the episodic memory traces. For instance, after going for a leisurely walk around a lake, we might only remember a path, trees and water, but not the types of trees, the shape of their leaves, or state of bloom. However, we are more likely to remember these details, if we had been given the task to study the trees

² I use the term “episodic memory trace” to refer to the quintessential information that is stored in the brain about a particular episode and “episodic memory” to refer to the compound phenomenon that involves the episodic memory trace, but also other components and processes.

around the lake. I propose that episodic memory includes only gist information, because it stores a semantic representation of the input, not the rich sensory input itself. The semantic representation is more abstract and combines various objects to compound entities. For instance, in most instances we represent a car as a single entity rather than a collection of tires, windows, doors and other parts. Note that each of the parts is in itself also a semantic representation. Some might object that possessing a semantic representation does not qualify as having a semantic memory and they might be right, but that is a different debate. One more important feature of semantic representations is that they are not only categorical, e.g. a woman, but can also represent particular objects or persons, e.g., my mother. My suggestion that semantic representations are stored in episodic memory is similar Teyler and DiScenna's (1986) suggestion that episodic memory stores indices of the items rather than the items themselves. Semantic representations are probably localized in sensory areas of the neocortex (Huth et al., 2012).

We cannot only represent objects and people, but also associate properties to them and store the relationships between them. This information I call semantic information. It can involve symbolic rules involving categories, e.g. “police cars are blue” or “policemen carry guns”, or idiosyncratic properties of and relationships between particular objects, as long as they are consistent across different episodes, such as, for example, the name of a particular person (Quiñero et al., 2005) or information of the kind “The Eiffel Tower is in Paris” or “my car is red”. As these examples indicate, semantic information is about items of which we have developed semantic representations. Semantic information are general facts that are usually extracted from multiple experiences and/ or apply to more than one instance (Maguire and Frith, 2004). This slow learning accords well with findings in computational studies that, during training, information has to be presented multiple times to a neural network interleaved with previously learned information (e.g., McClelland et al., 1995). The structure of semantic information can be quite different, e.g. hierarchical, sequential, or map-like. So its representational format must support these different structures. Since semantic information is also discovered slowly, the discovered structure might change over time. Therefore the

representation must be flexible and support changes in the organization of the information. The association areas of the neocortex are likely specialized for these purposes (Svoboda et al., 2006).

Sequential episodic memory traces

Episodic memories are stored incidentally without a special trigger (Tulving, 1972). In the vast majority of cases, the potential importance of a particular episodic memory is not discernible at the time of storage and the usefulness of the stored information might only become apparent in the future. Since episodes unfold in time, it has been suggested that episodes are sequences of events and that they are represented by sequences of neuronal patterns (Cheng, 2013; Werning and Cheng, 2014; Cheng and Werning, 2015). Several prior studies have used sequences to model episodic memory (Jensen and Lisman, 1996; Levy, 1996), but this idea has not been adopted widely in the field. I suggest that the episodic memory trace forms the backbone of episodic memory providing the sequential ordering of events to which the items involved in the events can be associated to. Since episodic memory traces are linked to specific, experienced episodes, the traces have to be stored rapidly after a single experience (Ennaceur and Delacour, 1988). The episodic memory trace is the quintessential component of episodic memory. While other components involved in episodic memory serve important functions in other cognitive processes, episodic memory traces exclusively contribute to episodic memory. The reverse is also true. Removing the episodic memory trace from the system, by removing the hippocampus, abolishes episodic memory.

The hippocampal circuitry appears to be specialized for rapidly storing sequences of neuronal patterns (Cheng, 2013). The core of this specialization might be the recurrent network in CA3 that is able to generate intrinsic sequences (Azizi et al., 2013), and the feedforward projections probably serve to store the associations between simultaneous inputs from the entorhinal cortex (Neher et al., 2015). Evidence from a number of species suggest that the hippocampus is involved in storing sequence memory, for instance in humans (Curran, 1997; Schendan et al., 2003), monkey (Kimble and Pribram, 1963), and rats (Agster et al., 2002; Fortin et al., 2002).

Intriguingly, hippocampal neurons generate different types of sequential activity intrinsically (Buhry et al., 2011). In the offline state, i.e., when the animal sits quietly or is asleep, neurons fire spikes in a temporal sequence of length between 50ms and 400ms (Lee and Wilson, 2002). These offline neuronal sequences correlate with the order, in which neurons were active at an earlier time in the online state (Nádasy et al., 1999; Mehta et al., 2002). It was therefore suggested that sequential neural activity in the offline state is a replay of sequential activity driven by prior experience. Replay has been studied mostly in rodent hippocampus, but has also been found during free recall of movie sequences in humans (Gelbard-Sagiv et al., 2008).

Scenario generation

The retrieval of episodic memories is not a passive process, in which information is simply found in its original form, but an active process, in which information about a past episode is (re-)generated (Dudai and Carruthers, 2005; Michaelian, 2011). We also call this mnemonic simulation (Cheng and Werning, 2015) or scenario generation (Cheng et al., 2016). During retrieval, a scenario is generated that more or less accurately represents a past episode. A scenario is based on the sequences and associations stored in the episodic memory traces. The gist information in the semantic representations can be expanded upon with details stored in the semantic system. Furthermore, semantic information might be used to fill in gaps or resolve inconsistencies in episodic memory, which might arise through inattention, encoding, storage, or retrieval errors (Cheng et al., 2016). These processes occurs automatically, even if subjects strive to accurately retrieve a past experience, and might account for the unreliable nature of human episodic memory (Loftus et al., 1978; Zaragoza and Lane, 1994; Loftus and Pickrell, 1995; Roediger and McDermott, 1995; Schacter, 2002). Scenario generation is different from the concept of “scene construction”, which Hassabis and Maguire (2007) introduced. The latter refers to scenes that are static spatial arrangements while scenarios extend in space as well as time. Another difference is that scenes are isolated entities, whereas scenarios can be embedded to allow constructions of larger narratives and reflection (Suddendorf, 2013).

Evidence for the component nature of episodic memory

Some evidence for the proposed component nature of episodic memory comes from developmental studies (Suddendorf and Redshaw, 2013). Even though adult-like episodic memory develops only by the age of about four in children (Nelson, 1993; Hayne and Imuta, 2011; Scarf et al., 2013), children as young as 2½ years old appear to store and retrieve episodic memory traces (Bauer et al., 1998; Wenner and Bauer, 1999). These latter studies found that children were able to re-enact a complex sequence of actions after observing it only once. Importantly, the sequence was arbitrary, and therefore the children could not use semantic information about the natural sequential order of events to reproduce it. If, however, there are so-called enabling relations between the elements of the sequence, even younger toddlers (in these studies, 16 months olds) could re-enact the sequence. One interpretation of these results is that children are first able to reproduce a sequence based on semantic information, then develop the ability to store and retrieve arbitrary sequences as episodic memory traces, and later develop the capacity to construct nested scenarios and to distill more complex semantic information. The final stages of the development of episodic memory continue well after age 4, when increases in working memory capacity, among others factors, lead to the construction of more complex episodic memories and larger narratives (Suddendorf and Redshaw, 2013).

Novel account of systems consolidation

Putting it all together, I propose the following account of systems consolidation. Episodic memory traces are the core component of episodic memory. They are represented as temporal sequences of neural activity that are stored only in the hippocampus. The hippocampal circuitry is optimized for rapidly storing associations between arbitrary items in temporal sequences. Structure hidden in specific episodic experiences can be extracted and stored as semantic information in the neocortex, which is optimized for discovering and representing such structures. This is consistent with observations that episodic memory greatly facilitates semantic learning (Gabrieli et al., 1988; Kitchener et al., 1998; Bayley et al., 2008). Since

semantic information can be extracted from episodic memory, it might be possible in some instances to retrieve the same information from both systems. This would be more likely for more remote memories, since there was more time for semantic learning. So it would then appear as if remote autobiographical information becomes independent of the hippocampus, because it can be retrieved as semantic information from neocortex (Cermak and O'Connor, 1983; Cermak, 1984; Kirwan et al., 2008). In this sense, systems consolidation is an epiphenomenon of slow semantic learning that is driven by episodic memory.

The passage of time is not the sole or decisive factor for the extraction of semantic information. Whether and when semantic information is extracted from episodic memory likely depends on a range of factors, including the number of retrievals (Cermak and O'Connor, 1983; Nadel and Moscovitch, 1998), the salience and importance of the information (Westmacott and Moscovitch, 2003; Westmacott et al., 2004) and the difficulty of extracting the information. This diversity of factors might account for the diversity in experimental findings. While some studies suggest that even memories of remote episodes never become independent of the hippocampus (Steinvorth et al., 2005), others find that they do (Kirwan et al., 2008).

Slow learning of semantic information does not imply that neocortical plasticity is in itself slow, only that discovering and/ or storing structured information takes time. Nor do I wish to imply that semantic information can only be learned from episodic memory. Plasticity in neocortex by itself is sufficient to support semantic learning (Vargha-Khadem et al., 1997; Sharon et al., 2011). Nevertheless, episodic memory might aid semantic learning for several reasons. First, stored experiences can be reactivated for interleaved learning (McClelland et al., 1995; Kali et al., 2004). Second, episodic memory traces can connect temporally discontinuous information (Staresina and Davachi, 2009; Pyka and Cheng, 2014). Third, the temporal structure of inputs can be exploited to extract semantic information (Wiskott and Sejnowski, 2002; Franzius et al., 2011). Since the first and third cases have been modeled and discussed more extensively in the literature, I focus on the second function in more detail below.

Learning associations across time gaps

We recently studied a model for learning an association between two temporally discontinuous pieces of information (Pyka and Cheng, 2014). Associations bridging a time gap occur, for instance, in trace conditioning, which has been found to depend on the hippocampus in human (McGlinchey-Berroth et al., 1997), rabbit (Solomon et al., 1986), rat (McEchron et al., 1998; Weiss et al., 1999), and mouse (Tseng et al., 2004). In addition, functional magnetic resonance imaging revealed increased hippocampal activation during trace as compared to delay conditioning (Cheng et al., 2008). In our model, we adopted a two layer neural network as a model of the cortico-hippocampal system. The key property of the model was that axonal conduction delays were heterogeneous, in particular, the delays between cortex and hippocampus were longer than delays within neocortex or within the hippocampus. If neocortex was isolated, it could not store associations between two patterns that were separated by more than 60 ms, because of the narrow time window imposed by spike-timing dependent plasticity. With the addition of the hippocampus, the combined network could store associations across a temporal separation of up to 140 ms. Intriguingly, after repeated retrievals of the association in the neocortico-hippocampal loop, neocortex alone was sufficient to retrieve the associations, i.e., the association was consolidated.

While semantic learning in general is likely to require much more sophisticated algorithms, this simple model offers at least two important insights. First, consolidation in the neocortico-hippocampal loop changes the information stored. One might think that the association was transferred from hippocampus to the neocortex, but actually it is rather an extraction of information, since the information about the temporal relationship, which might be an important aspect of episodic memory, is lost after consolidation. Second, systems consolidation in this model occurs as a byproduct of repeated retrieval in the absence of any dedicated consolidation mechanism.

Conclusion

I have proposed a new account of systems consolidation based on a recently proposed theory of the component-structure of episodic memory, in which episodic memory traces and semantic information are combined to generate scenarios during episodic memory retrieval. The key idea in this new account is that episodic memory traces are represented as somewhat rigid temporal sequences that can be flexibly associated with arbitrary items, while semantic information consists of flexible relationships between somewhat rigid items. These suggestions offer an explanation for the special role of the hippocampus in systems consolidation and are well supported by experimental evidence. To put the main points of this chapter succinctly: episodic memory traces are stored, episodic memory is generated, semantic information is learned, and systems consolidation is an epiphenomenon of semantic learning from episodic memory.

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